Second-Order Linear Homogeneous Equations

Section 1: Distinct Real Roots

1 Solve the auxiliary characteristic equation for $y'' - 5y' + 6y = 0$.

a. $r = 2$ and $r = 3$

b. $r = -2$ and $r = -3$

c. $r = 1$ and $r = 6$

d. $r = \pm\sqrt{6}$

2 What is the general solution to $y'' - 5y' + 6y = 0$?

a. $y = c_1 e^{2x} + c_2 e^{3x}$

b. $y = c_1 e^{-2x} + c_2 e^{-3x}$

c. $y = (c_1 + c_2 x)e^{2x}$

d. $y = c_1 \cos(2x) + c_2 \sin(3x)$

3 Solve the IVP $y'' - y = 0$ with $y(0) = 4$ and $y'(0) = 2$.

a. $y = 3e^{x} + e^{-x}$

b. $y = 2e^{x} + 2e^{-x}$

c. $y = 4e^{x}$

d. $y = e^{x} + 3e^{-x}$

Section 2: Repeated Real Roots

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4 What is the general solution to $y'' - 6y' + 9y = 0$?

a. $y = (c_1 + c_2 x)e^{3x}$

b. $y = c_1 e^{3x} + c_2 e^{-3x}$

c. $y = c_1 e^{3x} + c_2 x^2 e^{3x}$

d. $y = c_1 \cos(3x) + c_2 \sin(3x)$

5 Why is the factor x introduced into the second solution xe^{rx} for repeated roots?

a. To ensure linear independence from the first solution e^{rx}

b. Because the derivative of e^{rx} contains x

c. To make the solution satisfy $y(0) = 0$

d. By Euler's identity

6 Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = -1$.

a. $y = (2 + 3x)e^{-2x}$

b. $y = (2 - x)e^{-2x}$

c. $y = 2e^{-2x}$

d. $y = (2 + x)e^{-2x}$

Section 3: Complex Conjugate Roots

7 If the characteristic equation has complex roots $r = \alpha \pm i\beta$, what is the real general solution?

a. $y = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$

b. $y = c_1 e^{\alpha x} + c_2 e^{\beta x}$

c. $y = e^{\beta x}(c_1 \cos(\alpha x) + c_2 \sin(\alpha x))$

d. $y = c_1 \cos((\alpha + \beta)x)$

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8 Find the general solution to $y'' + 16y = 0$.

a. $y = c_1\cos(4x) + c_2\sin(4x)$

b. $y = c_1e^{4x} + c_2e^{-4x}$

c. $y = (c_1 + c_2x)e^{4x}$

d. $y = c_1\cos(16x) + c_2\sin(16x)$

9 Solve the characteristic equation for $y'' - 2y' + 5y = 0$.

a. $r = 1 \pm 2i$

b. $r = 2 \pm 4i$

c. $r = -1 \pm 2i$

d. $r = 1 \pm 4i$

Section 4: Harmonic Motion and Damping Regimes

10 A mechanical oscillator modeled by $my'' + cy' + ky = 0$ is underdamped when:

a. $c^2 - 4mk < 0$ (oscillatory motion with exponentially decaying amplitude)

b. $c^2 - 4mk = 0$ (critically damped)

c. $c^2 - 4mk > 0$ (overdamped, no oscillations)

d. $c = 0$ (undamped simple harmonic motion)

11 Which damping condition returns a displaced mass to equilibrium in the shortest time without oscillating?

a. Critical damping ($c^2 = 4mk$)

b. Overdamping ($c^2 > 4mk$)

c. Underdamping

d. Zero damping

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12 What is the natural frequency ω_0 of the undamped oscillator $y'' + 25y = 0$?

a. $\omega_0 = 5$ rad/s

b. $\omega_0 = 25$ rad/s

c. $\omega_0 = 10$ rad/s

d. $\omega_0 = 2.5$ rad/s

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